HDOC Day 42

Aim

To develop a menu-driven C program using separate user-defined functions to remove vowels from a string, find the first repeating lowercase alphabet, and check whether two strings are anagrams.

Algorithm

1. Start the program and display a menu of four choices.

2. For vowel removal, read a string and copy only non-vowel characters into the output.

3. For the first repeating lowercase alphabet, scan left to right and return the first lowercase character already seen.

4. For the anagram check, count the frequencies of letters in both strings (ignoring spaces and case), then compare counts.

5. Repeat the menu until the user selects Exit.

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
#define MAX 500

void readLine(char str[], int size) {
    if (fgets(str, size, stdin) != NULL)
        str[strcspn(str, "\n")] = '\0';
}

void removeVowels(const char input[], char output[]) {
    int i, j = 0;
    for (i = 0; input[i] != '\0'; i++) {
        char c = (char)tolower((unsigned char)input[i]);
        if (c != 'a' && c != 'e' && c != 'i' &&
            c != 'o' && c != 'u')
            output[j++] = input[i];
    }
    output[j] = '\0';
}

char firstRepeatingLowercase(const char str[]) {
    int seen[26] = {0}, i;
    for (i = 0; str[i] != '\0'; i++) {
        if (str[i] >= 'a' && str[i] <= 'z') {
            int index = str[i] - 'a';
            if (seen[index]) return str[i];
            seen[index] = 1;
        }
    }
    return '\0';
}

int areAnagrams(const char a[], const char b[]) {
    int count[26] = {0}, i;
    for (i = 0; a[i] != '\0'; i++) {
        unsigned char c = (unsigned char)a[i];
        if (isalpha(c)) count[tolower(c) - 'a']++;
```

```
    }
    for (i = 0; b[i] != '\0'; i++) {
        unsigned char c = (unsigned char)b[i];
        if (isalpha(c)) count[tolower(c) - 'a']--;
    }
    for (i = 0; i < 26; i++)
        if (count[i] != 0) return 0;
    return 1;
}

int main(void) {
    char first[MAX], second[MAX], result[MAX];
    char repeat;
    int choice;
    while (1) {
        printf("\n--- STRING OPERATIONS MENU ---\n");
        printf("1. Remove all vowels\n");
        printf("2. First repeating lowercase alphabet\n");
        printf("3. Check anagrams\n");
        printf("4. Exit\nEnter choice: ");
        if (scanf("%d", &choice) != 1) {
            printf("Invalid input.\n");
            break;
        }
        while (getchar() != '\n');
        switch (choice) {
            case 1:
                printf("Enter a string: ");
                readLine(first, MAX);
                removeVowels(first, result);
                printf("Without vowels: %s\n", result);
                break;
            case 2:
                printf("Enter a string: ");
                readLine(first, MAX);
                repeat = firstRepeatingLowercase(first);
                if (repeat)
                    printf("First repeating lowercase: %c\n", repeat);
                else
                    printf("No repeating lowercase alphabet.\n");
                break;
            case 3:
                printf("Enter first string: ");
                readLine(first, MAX);
                printf("Enter second string: ");
                readLine(second, MAX);
                printf(areAnagrams(first, second) ?
                       "Strings are anagrams.\n" :
                       "Strings are not anagrams.\n");
                break;
            case 4:
                printf("Exiting program.\n");
                return 0;
            default:
                printf("Invalid choice. Try again.\n");
        }
    }
    return 0;
}
```

Sample Output

```
--- STRING OPERATIONS MENU ---
1. Remove all vowels
2. First repeating lowercase alphabet
3. Check anagrams
```

```
4. Exit
Enter choice: 1
Enter a string: Education is important
Without vowels: dctn s mprtnt

Enter choice: 2
Enter a string: programming
First repeating lowercase: r

Enter choice: 3
Enter first string: Listen
Enter second string: Silent
Strings are anagrams.

Enter choice: 4
Exiting program.
```

Result

The menu-driven C program was written to perform all three string operations using separate functions. The sample cases demonstrate the expected results.